

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – DEBUGGING & OPTIMISATION TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Debugging and Optimisation task
Weightage:	45%	Total Marks:	100
CO2 Statement:	Apply integer and floating-point arithmetic algorithms including Booth's multiplication, restoring/nonrestoring division, and IEEE 754 standard operations to solve fixed-point and floating-point computational problems.		
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Section / Batch:	Slot-B / 2025	Date:	05-08-2023

Instructions to Students

- Identify arithmetic errors and performance bottlenecks in the given problem.
- Analyse the execution steps to locate the source of errors.
- Debug the algorithm by correcting logical and computational faults.
- Optimize the solution using appropriate arithmetic techniques or algorithms.
- Evaluate the optimized solution for correctness, accuracy, and efficiency.

QUESTIONS

1. A program performing signed integer addition using 2's complement produces incorrect results when adding large values (e.g., +120 and +50 in 8-bit representation). Debug the issue by analyzing the binary computation and identifying whether overflow has occurred. Propose a solution to correctly detect and handle overflow in such cases.
2. A processor using a ripple carry adder shows significant delay in executing arithmetic operations in a realtime system. Debug the performance bottleneck by examining how carry propagates through each bit. Suggest an optimized solution using a carry look-ahead adder, and explain how it reduces delay.
3. A multiplication unit implementing Booth's algorithm gives incorrect results for certain negative operands. Debug the step-by-step execution of the algorithm, focusing on bit-pair checking, arithmetic shifts, and sign extension. Identify the possible source of error and suggest corrections to ensure accurate signed multiplication.
4. An embedded system using the restoring division algorithm experiences inefficiency due to repeated restoration steps, leading to increased execution time. Debug the algorithm's workflow and identify where unnecessary operations occur. Propose an optimized approach using the non-restoring division method, explaining how it improves performance.
5. A scientific application using IEEE 754 single precision floating-point representation produces inaccurate results when adding very small and very large numbers. Debug the issue by analyzing exponent alignment, normalization, and rounding errors. Suggest optimization techniques, such as using double precision or algorithmic adjustments, to improve accuracy.

Code of Conduct

I Vishwa U (192511192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Identification of Errors / Bugs	20	Accurately identifies all errors and issues with complete understanding of the problem	Identifies most errors with minor omissions	Identifies limited errors; misses key issues	Unable to identify errors or misunderstands the problem
Debugging Approach & Analysis	20	Uses effective debugging techniques with clear analysis and root cause identification	Applies suitable debugging methods with minor gaps	Uses basic debugging methods with limited analysis	No systematic debugging approach followed
Correctness of Debugged Solution	25	Provides completely corrected solution with accurate functionality and expected output	Provides working solution with minor errors	Partially correct solution with limited functionality	Incorrect solution or fails to resolve errors
Optimization Techniques Applied	20	Applies efficient optimization methods to improve performance, memory usage and quality	Performs optimization with noticeable improvements	Limited optimization with minimal improvements	No optimization applied or inefficient implementation
Testing and Documentation	15	Performs complete testing with proper validation and clear documentation	Provides adequate testing and documentation with minor issues	Limited testing and incomplete documentation	No proper testing or documentation provided

Marks Evaluation:

Q. No	Identification of Errors / Bugs (20)	Debugging Approach & Analysis (20)	Correctness of Debugged Solution (25)	Optimization Techniques Applied (20)	Testing and Documentation (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 20	/ 25	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

Q1. Debugging Signed Integer Addition (2's Complement Overflow)

Solution

Given numbers:

- $+120 = 01111000_2$
- $+50 = 00110010_2$

Addition:

```
01111000 (+120)
+ 00110010 (+50)
-----
```

10101010

Result = **10101010₂**

In 2's complement, this represents:

Invert : 01010101

+1 01010110 = 86

So the result is **-86**, which is incorrect because:

$120 + 50 = 170$

The maximum value representable in an 8-bit signed integer is **+127**. Since **170 > 127**, **overflow occurs**.

Overflow Detection

Overflow occurs when:

- Two positive numbers produce a negative result.
- Two negative numbers produce a positive result.

It can also be detected by checking:

Carry into MSB $\neq$ Carry out of MSB

Solution

- Detect overflow using the overflow flag (V).
- Increase data width (e.g., 16-bit integers).
- Generate an exception or error message when overflow occurs.

Theory

In 2's complement representation, signed numbers have a limited range.

For 8 bits:

- Minimum = **-128**
- Maximum = **+127**

When an arithmetic result exceeds this range, overflow occurs, causing an incorrect signed result. Modern processors detect overflow using hardware logic and set an overflow flag so software can handle the error.

Q2. Ripple Carry Adder Delay and Carry Look-Ahead Adder

Solution

In a Ripple Carry Adder (RCA), each full adder waits for the carry from the previous stage.

Example:

Bit0 → Carry0

↓

Bit1 → Carry1

↓

Bit2 → Carry2

↓

Bit3 → Carry3

The carry ripples through every stage.

For an n-bit adder:

Delay $\propto$ Number of bits

This creates a performance bottleneck in real-time systems.

Optimized Solution: Carry Look-Ahead Adder (CLA)

CLA calculates carries simultaneously using:

Generate (G) = $A \times B$

Propagate (P) = $A \oplus B$

Carry equations:

$$C1 = G0 + P0C0$$

$$C2 = G1 + P1G0 + P1P0C0$$

...

All carries are generated in parallel, greatly reducing delay.

Theory

A Ripple Carry Adder is simple and requires less hardware, but its delay increases with the number of bits because carries propagate sequentially.

A Carry Look-Ahead Adder predicts carry values using generate and propagate signals, allowing parallel carry computation. This significantly improves speed and is widely used in high-performance processors.

Q3. Debugging Booth's Multiplication Algorithm

Solution

Possible reasons for incorrect multiplication of negative numbers:

1. Incorrect bit-pair checking (Q_0 and Q_{-1}).
2. Wrong arithmetic right shift.
3. Missing sign extension.
4. Incorrect initialization of Q_{-1} .

Booth's decision table:

Q_0 Q_{-1} Operation

00	No operation
01	$A = A + M$
10	$A = A - M$
11	No operation

After each operation:

- Perform **Arithmetic Right Shift** on **A**, **Q**, Q_{-1}
- Preserve the sign bit.

Corrections

- Correctly initialize $Q_{-1} = 0$.
- Preserve MSB during arithmetic shift.
- Extend sign bits properly.
- Verify subtraction uses 2's complement.

Theory

Booth's algorithm efficiently multiplies signed binary numbers by examining pairs of bits. It reduces the number of additions and subtractions for consecutive 1's.

Accurate multiplication depends on correct bit-pair evaluation, arithmetic right shifting with sign extension, and proper initialization. Errors in these steps mainly affect negative operands.

Q4. Restoring vs Non-Restoring Division

Solution

In Restoring Division:

1. Shift dividend.
2. Subtract divisor.
3. If result becomes negative:
 - Restore previous value by adding divisor again.
4. Continue.

Problem:

Many iterations require an additional restoration step.

Subtract

↓

Negative?

↓ Yes

Restore

↓

Next iteration

This increases execution time.

Optimized Solution: Non-Restoring Division

Instead of restoring immediately:

- If result is negative, perform addition in the next iteration.
- If result is positive, continue subtraction.

This eliminates repeated restoration operations.

Advantages:

- Fewer arithmetic operations.
- Faster execution.
- Better processor efficiency.

Theory

Restoring Division restores the partial remainder whenever subtraction produces a negative value, increasing hardware operations and delay.

Non-Restoring Division avoids immediate restoration by adjusting the next operation based on the sign of the previous remainder. This reduces execution time and improves performance in embedded systems.

Q5. IEEE 754 Floating Point Accuracy

Solution

Problem:

Adding a very large number and a very small number.

Example:

1.0×10^8

+

1.0

Steps:

1. Exponent Alignment

The smaller number's mantissa is shifted right repeatedly to match the larger exponent.

2. Loss of Precision

After many shifts:

000000...0001

The significant bits may disappear.

3. Addition

Only the large number remains significant.

4. Normalization

Result is normalized.

5. Rounding

Remaining bits are rounded according to IEEE 754 rules, introducing small errors.

Optimization

- Use **double precision (64-bit)** instead of single precision.
- Use compensated summation algorithms (e.g., **Kahan Summation**).
- Add numbers of similar magnitude first.
- Avoid repeated accumulation of very small values into very large values.

Theory

IEEE 754 single precision uses:

- **1-bit Sign**
- **8-bit Exponent**
- **23-bit Mantissa**

During floating-point addition, exponents must first be aligned. Large exponent differences force the smaller number to lose significant bits, causing rounding errors and reduced accuracy.

Using double precision or improved summation algorithms increases precision and minimizes numerical errors, making computations more reliable in scientific applications.